

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST

- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Answer:

1) Correct HTML tag for video embedding: Option B — <video src="video.mp4" controls></video>. The <video> element is the standard HTML5 tag specifically designed to embed video, is natively supported by all modern browsers, and accepts attributes like "controls" to show playback controls. <media> and <movie> are not real HTML elements, and <vid> is not a recognized tag either.

2) Appropriate HTTP method for form submission: Option B — POST. POST sends the submitted data in the request body rather than the URL, so sensitive registration details are not exposed in the browser history, bookmarks, or typical server logs. POST also supports larger and more complex payloads than GET, making it the conventional and more secure choice for submitting form data to the server.

3) Proper semantic element for navigation: Option B — <nav>. The <nav> element semantically marks a block of navigation links, which improves accessibility (screen readers can jump straight to it) and helps search engines understand page structure. A generic <div> carries no semantic meaning, so it does not communicate the section's purpose the way <nav> does.

4) Correct XML syntax: Option B —

<note><to>User</to><from>Admin</from></note>. Explanation of the errors in the other options: Option A is missing the closing </note> tag, so the document is not well-formed. Option C closes <to> with a mismatched </from> tag, breaking XML's rule that tags must close in the same order they were opened. Option D has an incomplete closing tag ("</note" without the closing angle bracket) and never

properly closes the root element. XML requires every element to have a matching, correctly nested closing tag, and only option B satisfies this.

5) Correct protocol for web communication: Option B — HTTP (used in practice as HTTPS for secure communication). HTTP is the application-layer protocol specifically designed for transferring web pages and resources between browsers and servers. FTP is used for file transfer, SMTP for sending email, and SNMP for managing network devices — none of these are used for standard web page communication.

Code of Conduct:

I (K.sai kumar/ 192372080) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

k.sai kumar

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand